FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°2_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES INFORMATIQUES

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L'objectif de ce TD est de:

1. Comprendre le principe d'une sauvegarde
2. Utiliser tar et rsync
3. Automatiser une sauvegarde avec cron
4. Vérifier et restaurer une sauvegarde
5. Mettre en place une sauvegarde vers un serveur distant
6. Utiliser rsync via SSH
7. Sécuriser la connexion par clé SSH
8. Restaurer les données

Exercice 1

1-Mettre en place une vraie stratégie de sauvegarde locale

2-Créer des sauvegardes datées

3-Automatiser avec cron

4-Mettre en place une rotation des sauvegardes

5-Tester une restauration

Créer le dossier de destination

```
lens@lens:~$ sudo rmdir /backup
lens@lens:~$ sudo rmdir /var/www/
lens@lens:~$ sudo mkdir /backup
lens@lens:~$ sudo mkdir -p /var/www/
lens@lens:~$ sudo mkdir -p /etc/
lens@lens:~$ sudo mkdir -p /home/devops/projects
lens@lens:~$
```

Créer le script de sauvegarde

```
lens@lens: ~
GNU nano 7.2                                        sauvegarde.sh
#!/bin/bash

# Dossier de destination
DEST="/backup"

# Date du jour
DATE=$(date +'%Y-%m-%d_%H-%M-%S')

# Sauvegarde de /var/www/
tar -czf $DEST/www_$DATE.tar.gz /var/www/

# Sauvegarde de /etc/
tar -czf $DEST/etc_$DATE.tar.gz /etc/

# Sauvegarde de /home/devops/projects
tar -czf $DEST/projects_$DATE.tar.gz /home/devops/projects

# Rotation : garder seulement les 7 dernières sauvegardes pour chaque type
ls -t $DEST/www_*.tar.gz | tail -n +8 | xargs rm -f
ls -t $DEST/etc_*.tar.gz | tail -n +8 | xargs rm -f
ls -t $DEST/projects_*.tar.gz | tail -n +8 | xargs rm -f

                                        [ Read 21 lines ]
^G Help       ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit       ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
lens@lens:~$ nano sauvegarde.sh
lens@lens:~$ chmod +x sauvegarde.sh
lens@lens:~$ sudo ./sauvegarde.sh
tar: Removing leading `/' from member names
tar: Removing leading `/' from member names
tar: Removing leading `/' from member names
lens@lens:~$ ls -lh /backup
total 1.8M
-rw-rw-r-- 1 lens lens 569K Mar 15 16:50 etc_2026-03-15_16-50-01.tar.gz
-rw-r--r-- 1 root root 577K Mar 15 16:52 etc_2026-03-15_16-52-25.tar.gz
-rw-r--r-- 1 root root 577K Mar 15 16:55 etc_2026-03-15_16-55-35.tar.gz
-rw-rw-r-- 1 lens lens  123 Mar 15 16:50 projects_2026-03-15_16-50-01.tar.gz
-rw-r--r-- 1 root root  123 Mar 15 16:52 projects_2026-03-15_16-52-25.tar.gz
-rw-r--r-- 1 root root  123 Mar 15 16:55 projects_2026-03-15_16-55-35.tar.gz
-rw-rw-r-- 1 lens lens  111 Mar 15 16:50 www_2026-03-15_16-50-01.tar.gz
-rw-r--r-- 1 root root  111 Mar 15 16:52 www_2026-03-15_16-52-25.tar.gz
-rw-r--r-- 1 root root  111 Mar 15 16:55 www_2026-03-15_16-55-35.tar.gz
lens@lens:~$ crontab -e
```

Automatiser avec cron

```
lens@lens: ~

  GNU nano 7.2                              /tmp/crontab.luxp3J/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /home/ton_user/sauvegarde.sh


                                                    [ Wrote 24 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/
```

```
lens@lens:~$ sudo crontab -e
crontab: installing new crontab
lens@lens:~$ sudo ./sauvegarde.sh
tar: Removing leading `/' from member names
tar: Removing leading `/' from member names
tar: Removing leading `/' from member names
lens@lens:~$ sudo ls /backup
etc_2026-03-15_16-50-01.tar.gz  projects_2026-03-15_16-50-01.tar.gz  www_2026-03-15_16-50-01.tar.gz
etc_2026-03-15_16-52-25.tar.gz  projects_2026-03-15_16-52-25.tar.gz  www_2026-03-15_16-52-25.tar.gz
etc_2026-03-15_16-55-35.tar.gz  projects_2026-03-15_16-55-35.tar.gz  www_2026-03-15_16-55-35.tar.gz
etc_2026-03-15_16-59-06.tar.gz  projects_2026-03-15_16-59-06.tar.gz  www_2026-03-15_16-59-06.tar.gz
etc_2026-03-15_16-59-32.tar.gz  projects_2026-03-15_16-59-32.tar.gz  www_2026-03-15_16-59-32.tar.gz
lens@lens:~$
```

Tester une restauration

```
lens@lens:~$ ls /backup
etc_2026-03-15_16-50-01.tar.gz  projects_2026-03-15_16-50-01.tar.gz  www_2026-03-15_16-50-01.tar.gz
etc_2026-03-15_16-52-25.tar.gz  projects_2026-03-15_16-52-25.tar.gz  www_2026-03-15_16-52-25.tar.gz
etc_2026-03-15_16-55-35.tar.gz  projects_2026-03-15_16-55-35.tar.gz  www_2026-03-15_16-55-35.tar.gz
etc_2026-03-15_16-59-06.tar.gz  projects_2026-03-15_16-59-06.tar.gz  www_2026-03-15_16-59-06.tar.gz
etc_2026-03-15_16-59-32.tar.gz  projects_2026-03-15_16-59-32.tar.gz  www_2026-03-15_16-59-32.tar.gz
lens@lens:~$ sudo tar -xzf /backup/etc_2026-03-15_16-50-01.tar.gz -C /
lens@lens:~$ client_loop: send disconnect: Connection reset

C:\Users\Lens Sandro Petiote>
```

Host Driver: Default
Controller: ICH AC97

```
lens@lens:~$ sudo crontab -l
[sudo] password for lens:
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /home/ton_user/sauvegarde.sh
```

Exercice 2

1-Mettre en place une sauvegarde distante sécurisée
2-Configurer une authentification SSH par clé
3-Automatiser avec cron
4-Restaurer des données en cas d’incident

Création de l’utilisateur dédié et du dossier de sauvegarde

```
lens@lens:~$ sudo adduser backupuserr
[sudo] password for lens:
info: Adding user `backupuserr' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `backupuserr' (1002) ...
info: Adding new user `backupuserr' (1002) with group `backupuserr (1002)' ...
info: Creating home directory `/home/backupuserr' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for backupuserr
Enter the new value, or press ENTER for the default
        Full Name []: sandro lens
        Room Number []: 5566566556
        Work Phone []: 566676677
        Home Phone []: 566676666
        Other []: 0000
Is the information correct? [Y/n] y
info: Adding new user `backupuserr' to supplemental / extra groups `users' ...
info: Adding user `backupuserr' to group `users' ...
lens@lens:~$ sudo mkdir -p /backup/devcloud
lens@lens:~$ sudo chown -R backupuserr:backupuserr /backup/devcloud
```

Authentification SSH par clé SSH

```
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lens/.ssh/id_ed25519
Your public key has been saved in /home/lens/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:W/5hVhyh0hShEFc0wYfpmWk7cPI5UpsmMmgnu8/tyAw lens@lens
The key's randomart image is:
+--[ED25519 256]--+
|        o..+O*.  |
|         o ++o.. |
|          o.o=.  |
|          o.O. . |
|       .S .B =o  |
|      + ++o X.   |
|     .E+.o.++o   |
|      .= o + .   |
|      .o*.o .    |
+----[SHA256]-----+
lens@lens:~$ cat .ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIATPKJ27Qt8Vjc3ZXQs1Nl/9Jqw/VEZuIsm20zQ+3KAG lens@lens
lens@lens:~$ ssh-copy-id lens@10.79.242.79
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/lens/.ssh/id_ed25519.pub"
The authenticity of host '10.79.242.79 (10.79.242.79)' can't be established.
ED25519 key fingerprint is SHA256:nODC/5aoIY4E9BhwtIzMjRr6IoPB7ncus2ADLZNPqBw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
lens@10.79.242.79's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'lens@10.79.242.79'"
and check to make sure that only the key(s) you wanted were added.

lens@lens:~$ ssh-copy-id lens@10.79.242.79
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/lens/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
lens@10.79.242.79's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'lens@10.79.242.79'"
and check to make sure that only the key(s) you wanted were added.
```

Script de sauvegarde avec rsync

```
lens@lens:~$ sudo nano /usr/local/bin/backup_prod.sh
lens@lens:~$ sudo chmod +x /usr/local/bin/backup_prod.sh
lens@lens:~$ /usr/local/bin/backup_prod.sh
lens@lens:~$ cat /var/log/backup_prod.log
```

```
backupuser@lens: ~

  GNU nano 7.2                          /usr/local/bin/backup_prod.sh *
#!/bin/bash

# Variables
SOURCE_PATHS="/var/www/ /etc/ /home/devops/projects"
DEST_SERVER="SRV-BACKUP"          # ou l'adresse IP, ex. 10.79.242.79
DEST_PATH="/backup/devcloud"
USER="backupuser"
LOGFILE="/var/log/backup_prod.log"
DATE=$(date +"%Y-%m-%d %H:%M:%S")

echo "[$DATE] Début de la sauvegarde" >> $LOGFILE

# Boucle sur chaque répertoire critique
for PATH in $SOURCE_PATHS; do
    if [ -d "$PATH" ]; then
        rsync -avz --delete -e ssh $PATH $USER@$DEST_SERVER:$DEST_PATH >> $LOGFILE 2>&1
        if [ $? -eq 0 ]; then
            echo "[$DATE] Sauvegarde réussie pour $PATH" >> $LOGFILE
        else
            echo "[$DATE] ERREUR lors de la sauvegarde de $PATH" >> $LOGFILE
        fi
    else
        echo "[$DATE] ATTENTION : $PATH n'existe pas sur SRV-PROD" >> $LOGFILE
    fi
done

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

```
backupuser@lens:~$ ls -l /backup/devcloud/
total 32
drwxr-xr-x 108 backupuserr backupuserr 4096 Mar 15 17:58 etc_2026-03-15_16-45-34
drwxr-xr-x 108 backupuserr backupuserr 4096 Mar 15 17:58 etc_2026-03-15_21-03-01
drwxrwxr-x   3 backupuserr backupuserr 4096 Mar 15 20:47 projects_2026-03-15_16-45-34
drwxrwxr-x   3 backupuserr backupuserr 4096 Mar 15 20:45 projects_2026-03-15_20-42-23
drwxrwxr-x   3 backupuserr backupuserr 4096 Mar 15 21:03 projects_2026-03-15_21-03-01
drwxrwxr-x   2 backupuserr backupuserr 4096 Mar 15 20:46 www_2026-03-15_16-45-34
drwxrwxr-x   2 backupuserr backupuserr 4096 Mar 15 20:43 www_2026-03-15_20-42-23
drwxrwxr-x   2 backupuserr backupuserr 4096 Mar 15 21:03 www_2026-03-15_21-03-01
```

Automatisation avec cron

```
lens@lens: ~
 GNU nano 7.2                                    /tmp/crontab.yS1o1x/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/backup_prod.sh


                                                    [ Wrote 24 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Loc
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go
```

Restauration des données

```
backupuserr@lens:~$ rsync -avz backupuser@10.79.242.79:/backup/devcloud/etc/ /etc/
backupuser@10.79.242.79's password:
receiving incremental file list

sent 20 bytes  received 51 bytes  15.78 bytes/sec
total size is 0  speedup is 0.00
```

Questions

1.Quelle est la différence entre tar et rsync ?
R- tar est un outil destiné à créer des archives compressées regroupant plusieurs fichiers en un seul, ce qui facilite leur stockage ou transfert. En revanche, rsync est conçu pour synchroniser des répertoires et ne copie que les fichiers modifiés, ce qui le rend particulièrement adapté aux sauvegardes incrémentielles et régulières.

2. Différence entre scp et rsync ?
R- scp permet de transférer des fichiers de manière sécurisée via SSH, mais il recopie systématiquement l'intégralité des données. rsync, également utilisable avec SSH, optimise le transfert en ne déplaçant que les différences entre source et destination, ce qui réduit considérablement le temps et la bande passante nécessaires.

3. Pourquoi automatiser les sauvegardes ?
R- L'automatisation des sauvegardes garantit la régularité des copies de sécurité, limite les risques liés aux oublis ou erreurs humaines et assure une meilleure protection des données en cas de panne ou d'incident. Elle constitue une mesure proactive de continuité et de fiabilité.

4. Où sont stockées les tâches cron de l'utilisateur ?
R- Les tâches planifiées par un utilisateur via cron sont enregistrées dans le répertoire /var/spool/cron/crontabs/, avec un fichier propre à chaque utilisateur. Elles peuvent être consultées ou modifiées directement avec la commande crontab -e.

5. Comment planifier une sauvegarde tous les jours à 23h ?
R- Pour planifier une sauvegarde tous les jours à 23h, on doit ajouter une ligne dans la crontab avec la bonne syntaxe .

Coclusion.

Ce TD m'a permis de comprendre le principe d'une sauvegarde, automatiser,vérifier et restaurer une sauvegarde, mettre en place une sauvegarde vers un serveur distant, utiliser rsync via SSH, sécuriser la connexion par clé SSH, restaurer les données, donc la tache a été réussie.